

Suggested Solutions to Exercises 11.

The Black-Scholes formula for the price of a European call option under the standard assumptions, with strike price K and time to expiry T , is

$$S_0 N(d_1) - K e^{-rT} N(d_2),$$

where $N(\cdot)$ denotes the cumulative distribution function of a standard Normal, and

$$d_1 = \frac{\log(\frac{S_0}{K}) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}$$
$$d_2 = \frac{\log(\frac{S_0}{K}) + \left(r - \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}} = d_1 - \sigma \sqrt{T}$$

The Black-Scholes formula for the price of a European put option with strike price K and time to expiry T , is

$$K e^{-rT} N(-d_2) - S_0 N(-d_1),$$

where the notation is the same as above.

1. (a) We have that:

$$d_1 = \frac{\log(49/50) + (0.05 + 0.2^2/2) \times 0.384615}{0.2\sqrt{0.384615}} = 0.05418$$

and

$$d_2 = \frac{\log(49/50) + (0.05 - 0.2^2/2) \times 0.384615}{0.2\sqrt{0.384615}} = -0.06985.$$

The call option price for one share is

$$49N(0.05418) - 50e^{-0.05 \times 0.384615}N(-0.06985) = 2.4005.$$

- (b) The put option price for one share is

$$50e^{-0.05 \times 0.384615}N(-0.06985) - 49N(-0.05418) = 2.4482.$$

- (c) $2.4005 - 2.4482 = -0.0477$ and

$$S - Ke^{-rT} = 49 - 50e^{-0.05 \times 0.384615} = -0.048.$$

2. Here $S = 110$, $r = 3\%$, $T = 2$ and $\sigma = 0.07$. If S_2 is the stock price after two years, a financial product pays $S_2 - 120$ if $S_2 > 120$, $100 - S_2$ if $S_2 < 100$ and zero otherwise.

This is equivalent to a portfolio of 2 two-year European options, precisely 1 long call with strike $X_c = 120$ and 1 long put with strike $X_p = 100$. Therefore the value of the product equals the value of the portfolio, i.e. $f = f_{c120} + f_{p100}$. We use Black-Scholes to find the two values to sum.

Call at 120: The two constants are $d_1 = -0.233$ and $d_2 = -0.32235$. So the option value is $\text{Call} = 3.0961$.

Put at 100: The two constants are $d_1 = 1.618$ and $d_2 = 1.5194$. So the value of the option is $\text{Put} = 0.2695$.

The value of the financial product is

$$\text{Call} + \text{Put} = 3.0961 + 0.2695 = 3.3656.$$

3. (a) The derivative is a long call option at strike price $K = 4$.

- (b) The derivative is composed of a binary option (which pays 0.2 if $S_T \leq 4$ and 0 otherwise) and a long bull spread (long a call option at strike price $K_1 = 4$ and short another call option at strike price $K_2 = 5$).

We can use the Black-Scholes formulas to price the bull spread option. For the binary option, we use risk-neutral valuation and we have, similarly to question 4 below,

$$\begin{aligned} f &= e^{-rT} \mathbb{E} [f_T | S_0] = 0.2 \mathbb{P} [S_T \leq K] \\ &= 0.2 \mathbb{P} \left[\frac{B_T}{\sqrt{T}} < \frac{\log(K/S) - (r - \sigma^2/2)T}{\sigma\sqrt{T}} \right] \\ &= 0.2 N(-d_2). \end{aligned}$$

4. (a) We need to find

$$\begin{aligned} \mathbb{P} [S_T > K] &= \mathbb{P} [S \exp(\sigma B_T + (\mu - \sigma^2/2)T) > K] \\ &= \mathbb{P} [\sigma B_T + (\mu - \sigma^2/2)T > \log(K/S)] \\ &= \mathbb{P} \left[\frac{B_T}{\sqrt{T}} > \frac{\log(K/S) - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}} \right] \\ &= 1 - \mathbb{P} \left[\frac{B_T}{\sqrt{T}} < \frac{\log(K/S) - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}} \right]. \end{aligned}$$

Since $\mu = 0.16$, $\sigma = 0.35$, $K = 40$, $S = 38$, $T = 1/2$, we have

$$\mathbb{P} [S_T > K] = 1 - N(0.00775) = 0.4969.$$

Using the Black-Scholes formula, the price of this option is 4.25.

- (b) The probability that the put option is exercised is

$$1 - 0.4969 = 0.5031$$

(since with probability 1, $S_T > K$ or $S_T < K$). Using put-call parity, the price of this option is 3.18.

5. We have that:

- $X_0 = 0$, and $t \mapsto X_t$ has continuous sample paths.

– For $s < t$, we have:

$$\begin{aligned}\text{cov}(X_t, X_s) &= \text{cov}(B_{2t} - B_t, B_{2s} - B_s) \\ &= \text{cov}(B_{2t}, B_{2s}) + \text{cov}(B_t, B_s) - \text{cov}(B_t, B_{2s}) - \text{cov}(B_{2t}, B_s) \\ &= 2s + s - \min\{t, 2s\} - s = 2s - \min\{t, 2s\}\end{aligned}$$

So, if $\min\{t, 2s\} = 2s$, then $\text{cov}(X_t, X_s) = 0$.

This is not a standard Brownian motion!

6. (a) - $X_0 = 0$, and $t \mapsto X_t$ has continuous sample paths.
 - For $s < t$, we have:

$$\text{cov}(X_t, X_s) = \frac{1}{4}\text{cov}(B_t, B_s)$$

- Increments on non-overlapping time intervals are independent.
 - For $s < t$, $X_t - X_s \sim N(0, \frac{1}{4}(t - s))$

So, this is a Brownian motion, with drift 0 and volatility $\sigma = 1/2$.

- (b) We have that:

$$\begin{aligned}\text{cov}(B_{3t} - tB_{2t}, B_s) &= \text{cov}(B_{3t}, B_s) - t\text{cov}(B_{2t}, B_s) \\ &= s - ts = s(1 - t).\end{aligned}$$

- (c) We have that:

$$\begin{aligned}\mathbb{E}[e^{-t/2}\sinh(B_t)|B_s] &= \mathbb{E}[e^{-t/2}\frac{e^{B_t}-e^{-B_t}}{2}|B_s] \\ &= \frac{e^{-t/2}}{2}(\mathbb{E}[e^{B_t}|B_s] - \mathbb{E}[e^{-B_t}|B_s]) \\ &= \frac{e^{-t/2}}{2}(e^{B_s+\frac{1}{2}(t-s)} - e^{-B_s+\frac{1}{2}(t-s)}) \\ &= e^{-s/2}\sinh(B_s).\end{aligned}$$

7. (a) Parameter r is called the speed reversion, θ is the long-run mean, and σ is a volatility parameter.

(b) For general function $f(\cdot)$, application of Itô's formula gives:

$$\begin{aligned} dY_t &= f'(X_t)dX_t + \frac{1}{2}f''(X_t)(dX_t)^2 \\ &= \left\{ r(\theta - X_t)f'(X_t) + \frac{\sigma^2 X_t}{2}f''(X_t) \right\} dt + \sigma\sqrt{X_t}f'(X_t)dB_t \end{aligned}$$

So, we need to have:

$$f'(x) = \frac{1}{\sigma\sqrt{x}}$$

We can choose:

$$f(x) = \frac{2}{\sigma}\sqrt{x}.$$